FACULTY OF COMPUTING
UTM Johor Bahru

Project : Phase 3

<Minisite.com>

SECD2523 - SYSTEM ANALYSIS AND DESIGN

SEMESTER I, SESSION 2022/2023

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1.0 Overview of Project

Due to the Covid-19 pandemic worldwide, especially in Malaysia, online shopping has become the main point of Malaysians' daily lives. Most of the sellers have been forced to turn their business online due to the restrictions on the operation which is a step to break the chain of the spread of Covid-19 by our government. Malaysians can shop online to get what is needed during the Movement Control Order restrictions. There is super convenient and safe for the public, whether seller or buyer. But it is very convenient for sellers to help them sell their products and gain income during the pandemic.

Mdm. Chelsea is a seller that had faced the pandemic and also going into the online business. Richmine Trading is one of the online businesses in Penang, covering areas in the whole of Malaysia no matter West Malaysia or East Malaysia. Richmine Trading provides the computer accessories such as Random Access Memory (RAM), Graphic Cards, and Storage Disk, which is an important things for people who can be running a business or having a remote lecture. Besides that, Richmine Trading also helps to build up the CPU upon the request from the buyer.

In this project, we will outline Mdm. Chelsea's current business system as well as her updated business system. The AS-IS system, which represents the information flow between the seller system and external organisations like the seller and banks, is used to illustrate the concept using logical data flow diagrams (DFD). We will also discuss the existing business procedures and processes at Richmine Trading.

The Logical DFD TO-BE system is developed to improve the AS-IS system based on the previously existing business process and workflow. The user experience and workload for Richmine Trading have both been improved by enhancements made to the logical DFD TO-BE system. The TO-BE system keeps track of all customer and order information, and an order analysis will be provided. To enhance the ordering experience for customers on Richmine Trading, additional elements such as menu availability, delivery person status, and feedback will be implemented. Mdm Chelsea will be able to change and raise the calibre of the food and services provided based on the data and analyses. The Physical DFD TO-BE system, Logical DFD TO-BE system, and Process Specification will be developed in order to convert the requirements into comprehensive and accurate system design specifications.

2.0 Problem Statement

1. **Hard to analyse order data and product status.**

    The stakeholder, Mrs. Chelsea needs to know the current status to update manually the order data record, she was facing problems and felt hard because she always spent her a lot of time doing analysis based on the sales. Besides that, she also found it hard to compare the sales based on the order data and the price with other companies due to the unsorted order data not being organised. Due to the inadequate analysed order data supply on time, she always missed catching up with the business timing since she won't know which items were popular and in people's demand by predicting through the data.

0. Hard to track the parcel that shipped out.

Mrs. Chelsea was having difficulties in tracking where the parcel goes and hard to estimate the time when the parcel reaches the customer since she was responsible for informing the customers regarding the product's information when the orders had been made successfully. Sometimes customers may place orders in advance and set to receive the deliveries at a specific time send to a particular place, thus Mrs. Chelsea and customers are unable to know where the parcel is located exactly and when the parcel will be shipped out, which courier services were responsible in shipping the parcel and even the time when the parcel arrives. Besides that, while tracking the parcel, the stakeholder also will need to know the order details, which can allow the user to determine the detail of the product's order, including the product purchased, order ID, and shipping address. Since she could not know track which product numbers were sold, in transit, or on delivery, it brings her trouble indirectly in analyzing the order of the product and the stock available.

0. **Hard to check and update the stock of products.**

Mrs. Chelsea was required to check the stock and inventory manually every day to ensure that there is a sufficient available stock that can be supplied on time to her business. While checking the stock, she found that it was hard to count the quantity of the product that still owns currently, but the problem is that she did not have enough time to always check the product stocks and the status one by one because of her limited ways must be done manually. In addition, Mrs. Chelsea was hard to update the product descriptions for more detailed information because she needed to update them one by one manually and repeat the process every time since there were too many product descriptions that needed to mention for each type of product, for example, retail price, the available stock, and product information.

Sometimes, she needs to check and predict the stock in advance and ensure a sufficient stock to allow the customers to place orders because some customers will add particular items to their carts before making the payment. Thus, updating the available stock information on time was hard on the online platforms. It was hard to know the stock listing since it is hard to list all the products published on online platforms and the details of products, including single products and bundle products.

0. **Hard to check the order according to the term.**

The problem causes her to have difficulties in analysing the order since when she needs to analyse the order data based on the product, she found that she needs to search for the specific product from the unsorted data, but she was unable to do so. Not only this, but when she wants to deliver the order, she needs to view the previous history order, she was unable to select to filter out all the unrequired data. Due to the limitations, she was hard and inefficient in completing her work because it always causes her to spend a lot of time searching for the product.

0. Hard to check the online payment status.

The problem causes her to have difficulties in analysing the order since when she needs to analyse the order data based on the product, she found that she needs to search for the specific product from the unsorted data, but she was unable to do so. Not only this, but when she wants to deliver the order, she needs to view the previous history order, she was unable to select to filter out all the unrequired data. Due to the limitations, she was hard and inefficient in completing her work because it always causes her to spend a lot of time searching for the product.

3.0 Proposed Solution

Minisite is a new system solution that we are trying to propose and introduce to the stakeholder, Mrs. Chelsea, in which we are concentrating on bringing convenience to the business. It is a system that provides a way to analyse the order data which make analysis processes to be easier because it will collect the data in organized form and list them based on the various type of products in terms of product ID, product type, brand, available stock, and so forth. This system will categorize the product based on the product status, defective, quantity, product description, and so forth. This system also provides an easy and free navigation method interface.

This system will provide airway bill printing, allowing users to print the airway bill according to the order and the customers can be tracked easily based on the tracking code provided by the courier services. Mrs. Chelsea able to know which number of product types with the specific number that will be shipped out previously, pending for shipping, and the product that will be shipping in the future. For example, the type of product that had been chosen by the customer will be stated along with the product numbers and delivery address, the date and the time of the product being shipped out and reaching the destination, and customer information clearly to help her identify her products has been successfully delivered to the customers.

Previously, it was hard to know the latest status of the product since she needs to update all the records manually. Thus, it was hard to synchronize all the order data and update it immediately. This system tried to provide and show the latest information to the stakeholder using the technique and the analyzing order's data method. This system will always update the stock available on time once the quantity of the product changes and the status of the product, thus it will increase the efficiency in saving time in checking and counting the product from numerous product statuses.

Besides that, this system will also provide Mrs. Chelsea with the product listing which state all the available product stock in a neat way which aims to help her in checking the stock and updating the stock. A sufficient organized order data supply could help the stakeholder catch up on the chance and predict the trend based on updating of the order data and the stock earlier. In addition, this system provides stock management for the user, enabling the stakeholder to make a stock adjustment. It provides a stock listing to list all the products published on online platforms and the details of the products.

Furthermore, Minisite was able to provide a way for the stakeholder Mrs. Chelsea to check the order according to their preferences. Firstly, this system required the stakeholder, Mrs. Chelsea, to log in advance and customize the data according to their interests and preferences. Hence, all the product data collected will be analysed, organized easily, and listed in the proper form based on the products. The customizing method allows the stakeholder to know where the data is located easily and how the data is being categorized. By searching the term, this system will show all the related data according to the term to allow her to select and filter out all the unrequired data to make the process easier. Not only this, but for user-friendly consideration, this system is designed to allow the user to select the date, either based on the period or the specific date to show the recent history of product data easily.

Moreover, this system provides a function of notification that can help stakeholders to be alert when there is any transformation or the payment had been made successfully by sending an email to inform Mrs. Chelsea. For example, a message regarding the defective product should be stated clearly along with the product ID to indicate the particular product exactly and a notification regarding the amount of payment that had been made by the customer successfully. Thus, she can easily identify the defective product to avoid being delivered to the customers, which lowers the business rating and avoids miss sending the products to the customer. Hence, this system can help in avoiding checking manually whether the payment was made successfully or not by the customers while using the online transfer method since this system will update immediately and notify them once the customer has made the payment successfully, along with a clear product description. Last but not least, when the product's stock was reaching the alert value, this system will try to send a notification to inform the stock status.

4.0 Requirement Analysis

4.1 Current business process (scenarios, workflow)

Here are the scenarios and workflow of current business process for stakeholder:

1. Login to the system.

    0. Customizing according to preference.

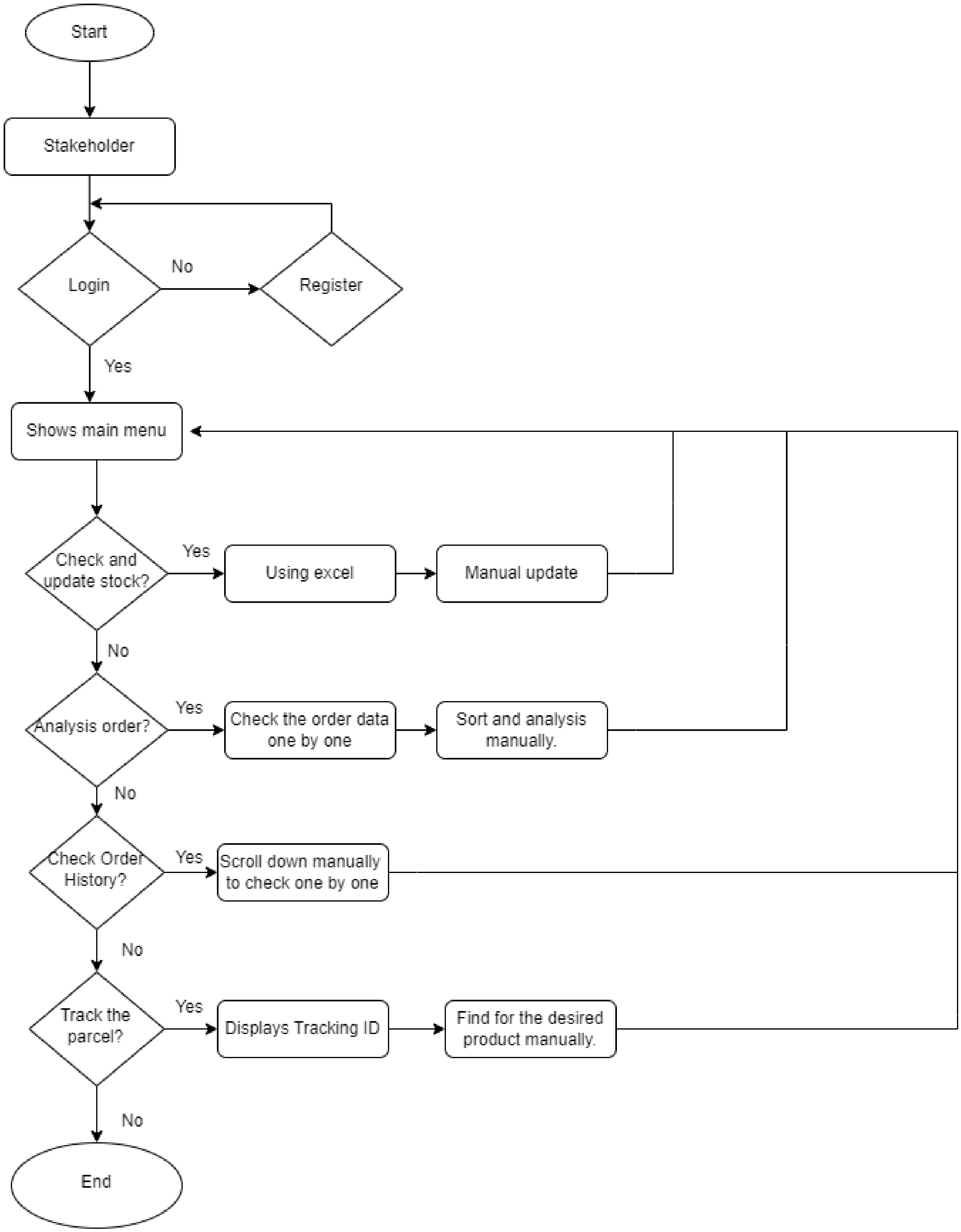
0. Some options are displayed on the main menu.

    0. Analysis Order.
        1. Unable to sort by preferences.
        2. Sort and analyze manually.

0. Track the parcel.
    1. Show Tracking ID only.
    2. Find the desired product tracking ID manually.

0. Check and update the stock.
    1. Using excel.
    2. Manual update.

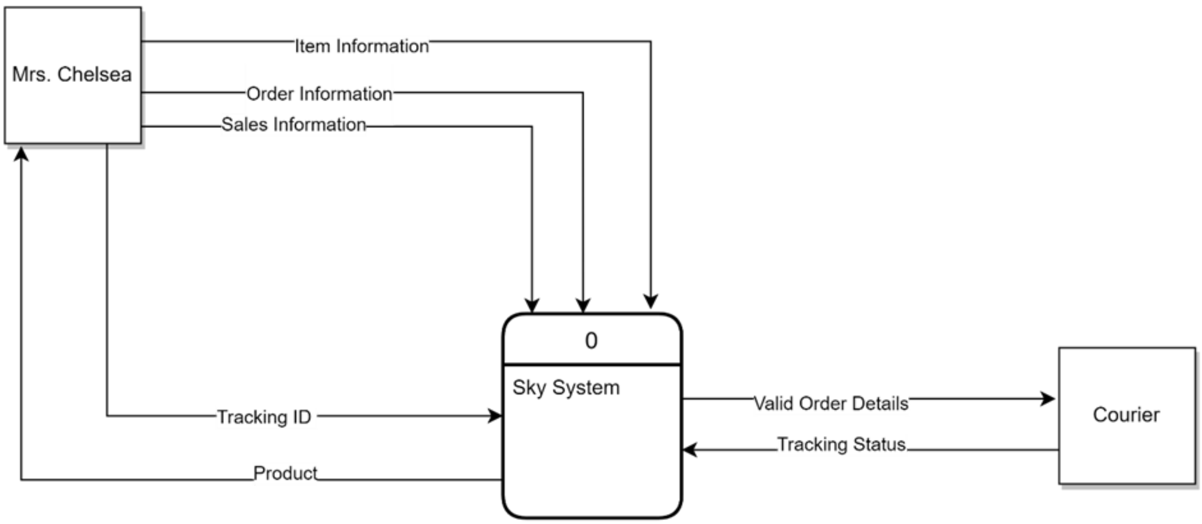
0. Option for checking Order History
    1. Scroll down manually to check one by one.



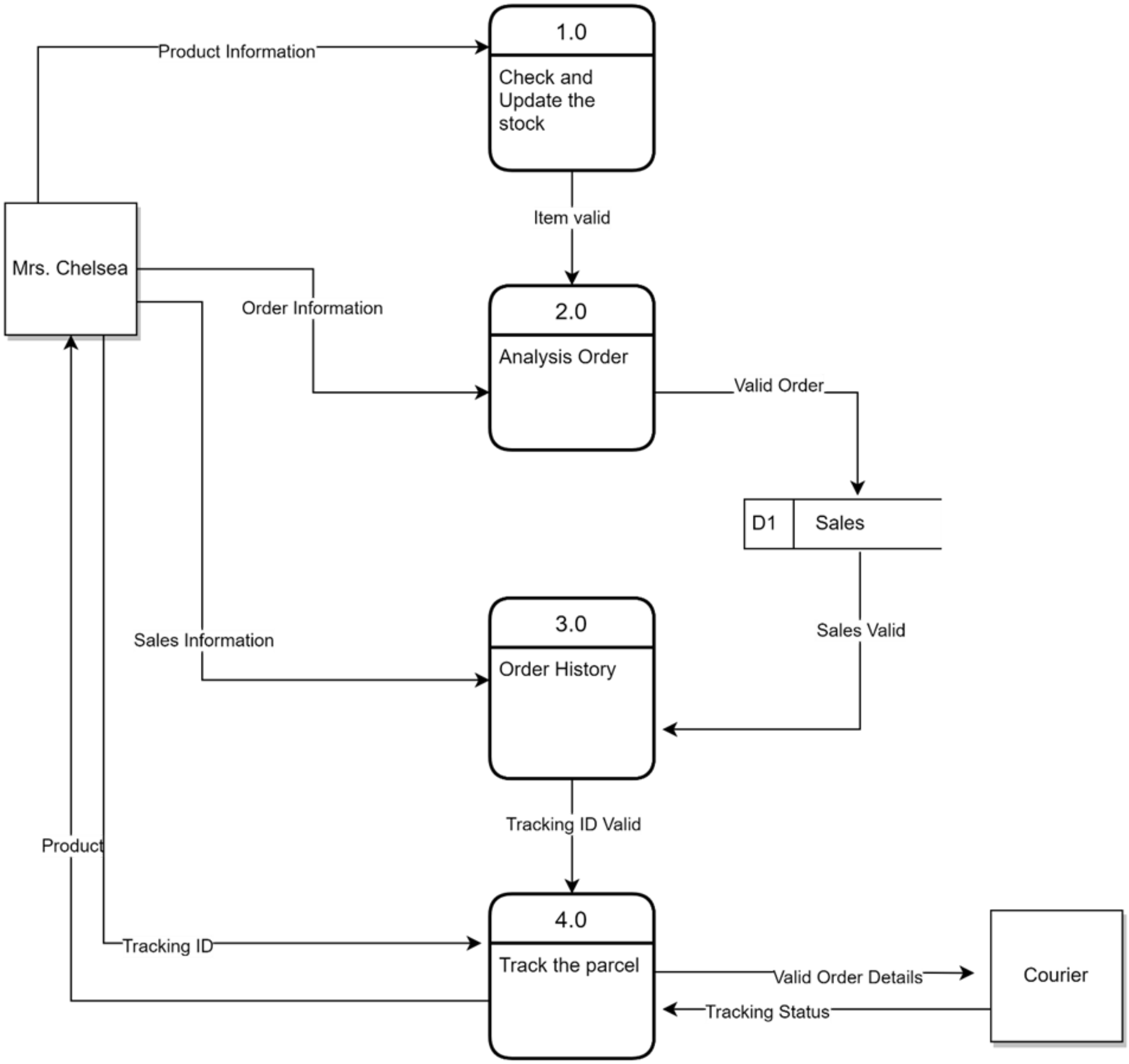
AS-IS System Workflow

5.0 Logical DFD for AS-IS System

5.1 Context Diagram

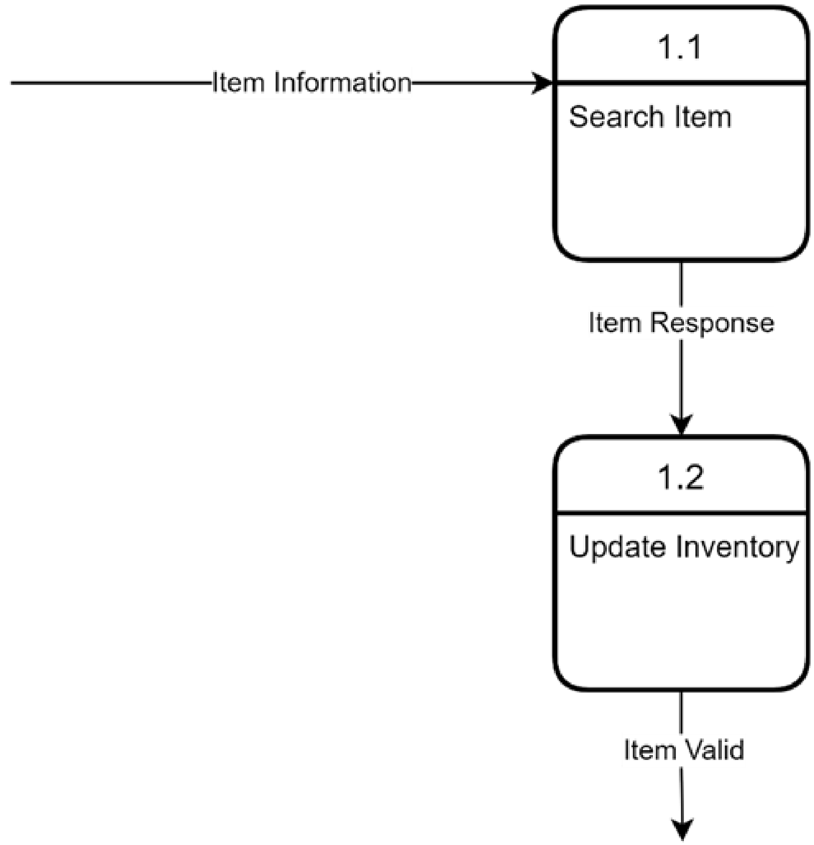


5.2 Level 0 Diagram

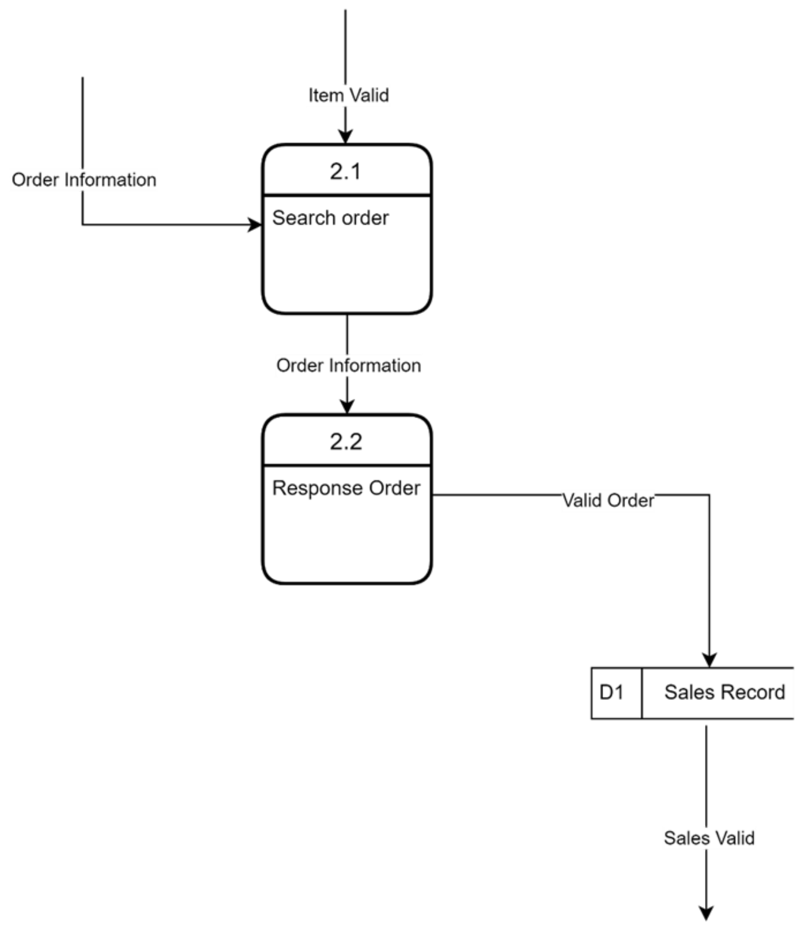


5.3 Level 1 Diagram

5.3.1 Process 1: Check and update the stock



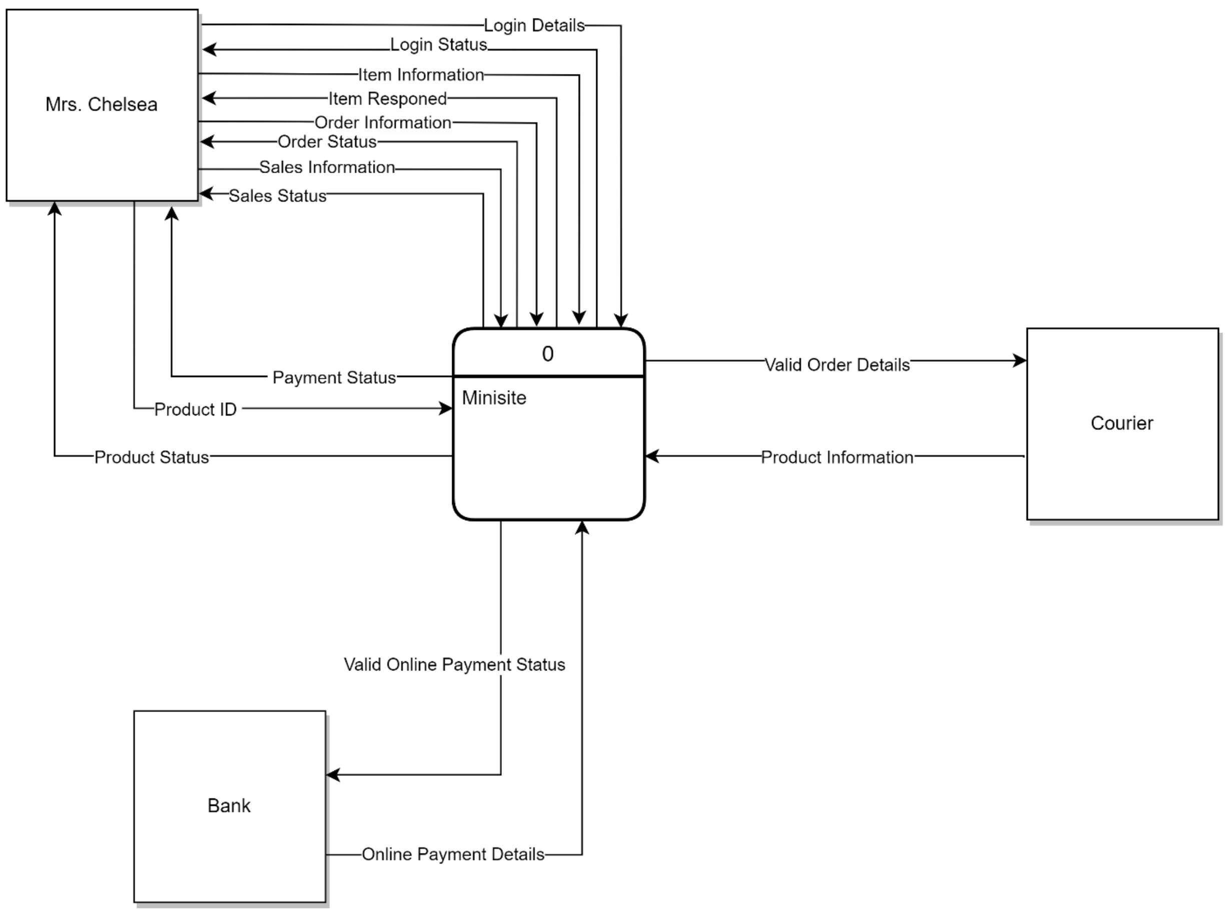
5.3.2 Process 2: Analysis of Order



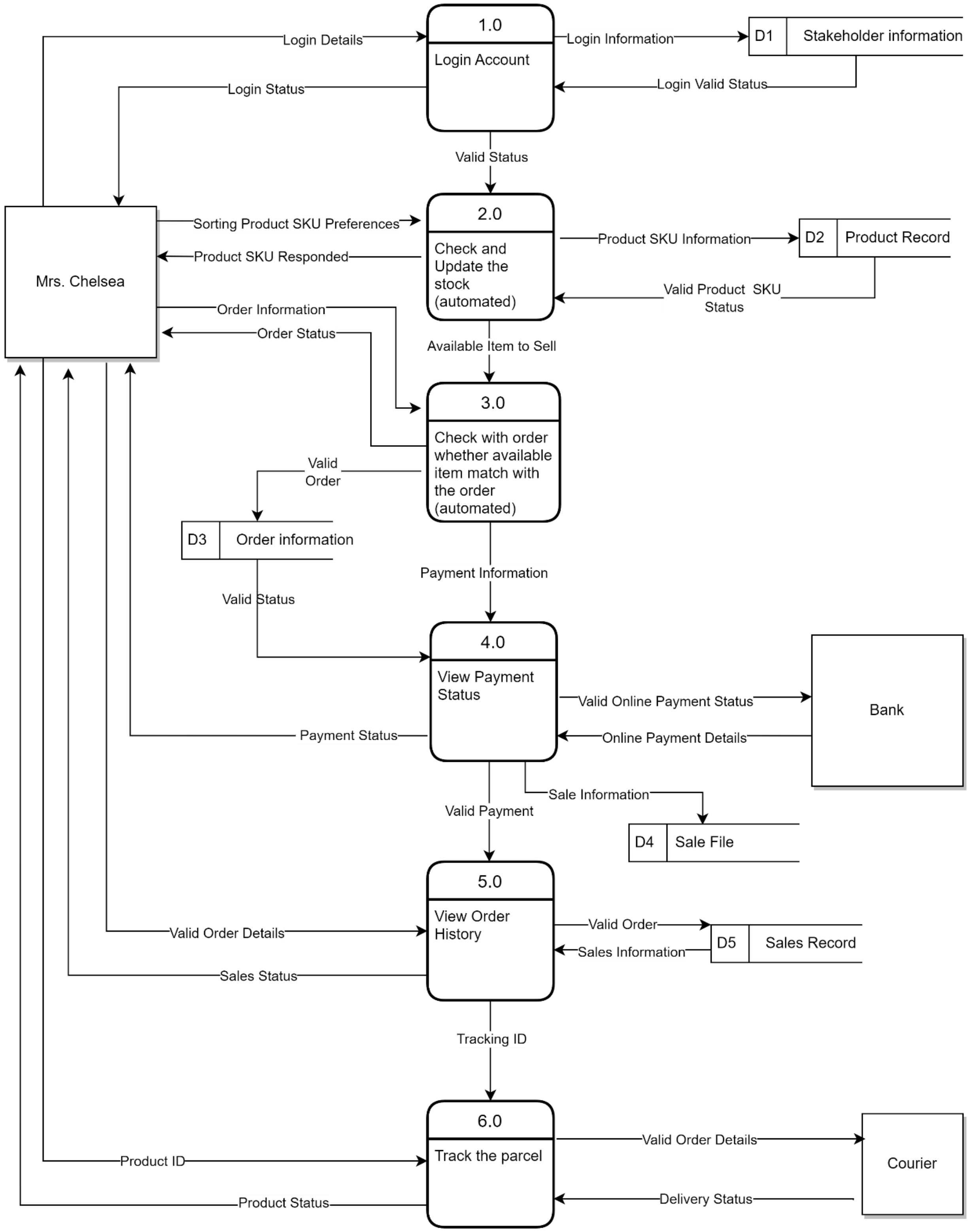
6.0 System Analysis and Specification

6.1 Logical DFD TO-BE system

6.1.1 Context Diagram

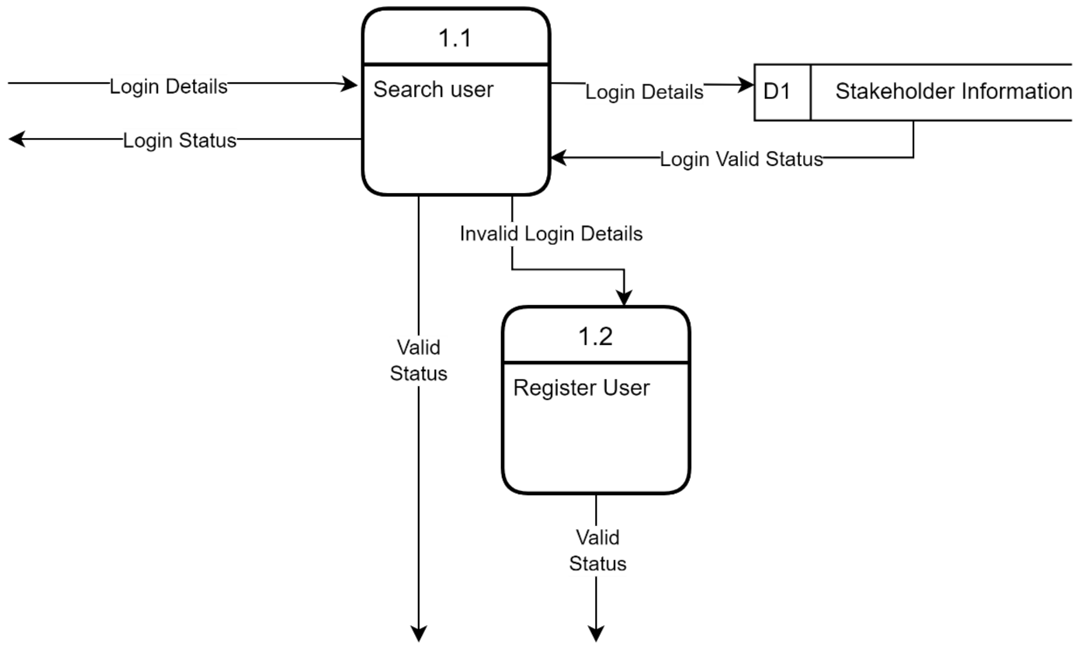


6.1.2 Level 0 Diagram

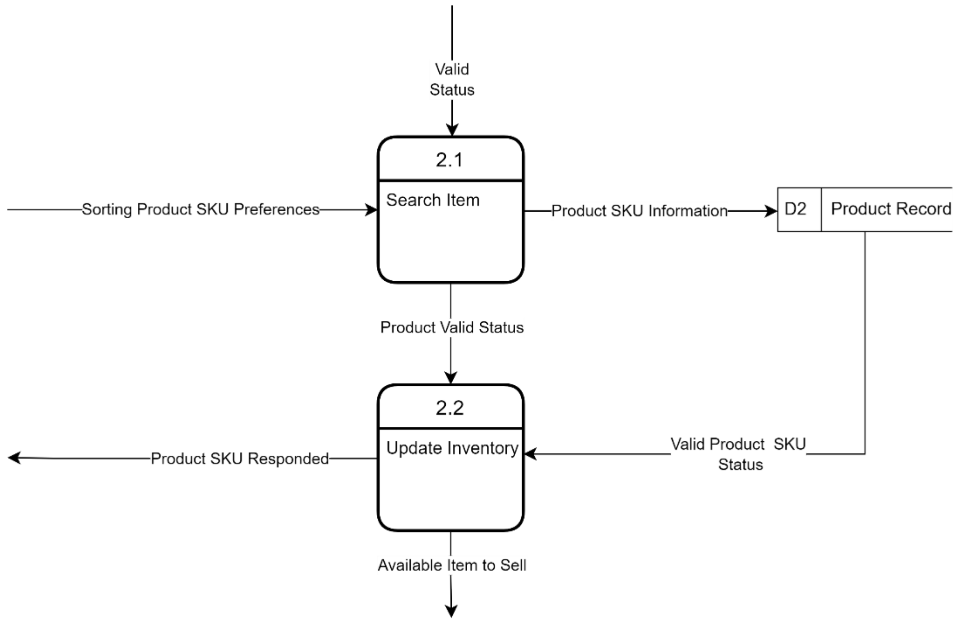


6.1.3 Level 1 Diagram

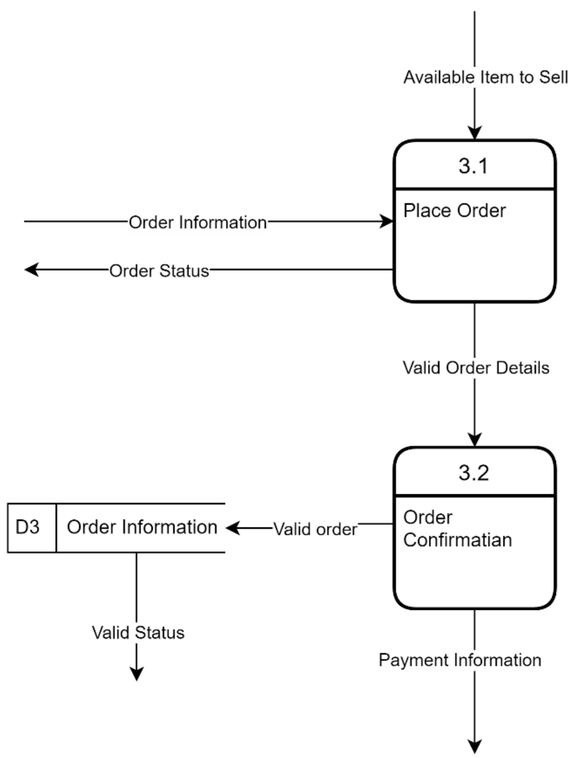
6.1.3.1 Process 1: Login Account



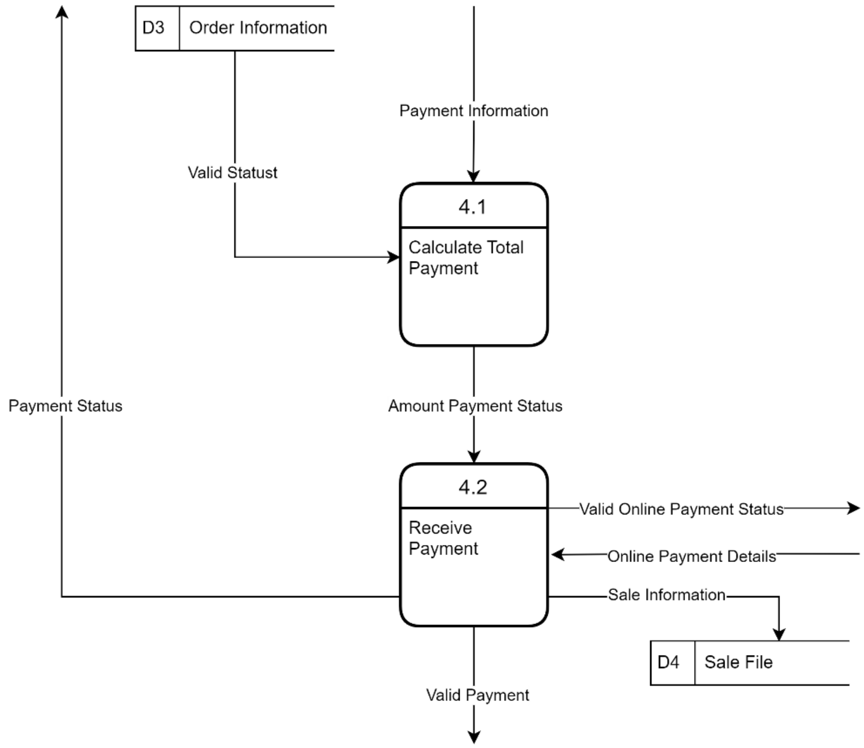
6.1.3.2 Process 2: Check and Update the Stock



6.1.3.3 Process 3: Check with order whether available item match with the order



6.1.3.4 Process 4: View Payment Status



6.2 Process Specification
Structured English is used to model and illustrate the process of the Logical TO-BE system based on the Logical DFD TO-BE.

6.2.1 Order History
DO

READ payment status

BEGIN IF

IF received the valid payment

  Read the sales information from database

  Read the valid order details from stakeholder

  THEN update the sales report based on valid payment and order information

  Store the valid order into database

ELSE continue

IF receive sale information

  PERFORM the generation of total sales report

  Display the Product ID

  Display the Tracking ID

DISPLAY the sales status to the stakeholder

END IF

6.2.2 Track the parcel
DO

Courier receive the valid order details from database

BEGIN IF

IF get tracking ID of the product

  Read product ID from stakeholder

  Get the customer information from courier

  PERFORM matching PRODUCT ID and customer information

  Display the valid delivery order details to the courier.

ELSE continue

IF deliver status received

  Read tracking status and estimation time from the courier

  PERFORM updating of the product’s status.

  Display the Product Status to the stakeholder

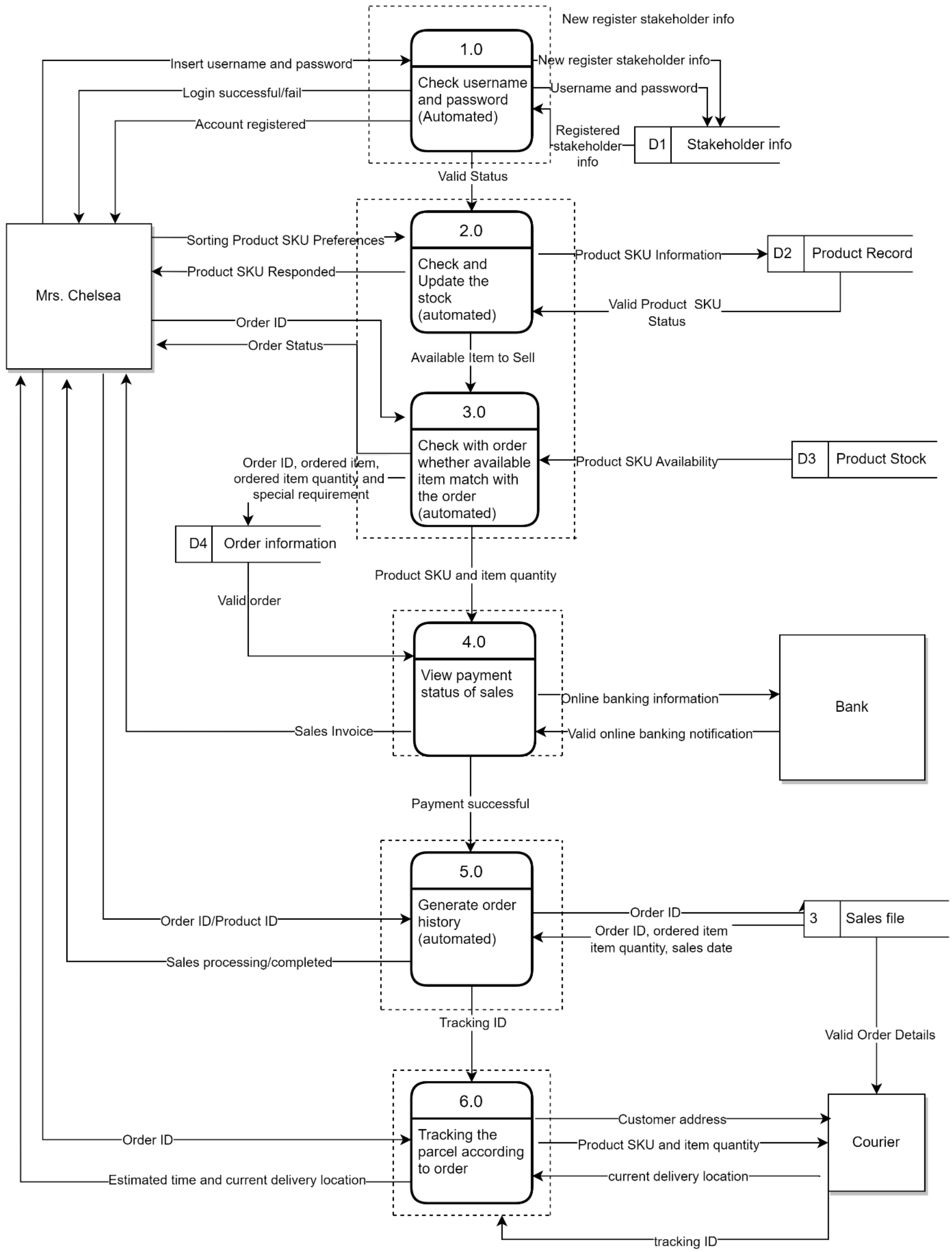
END IF

END

7.0 Physical System Design

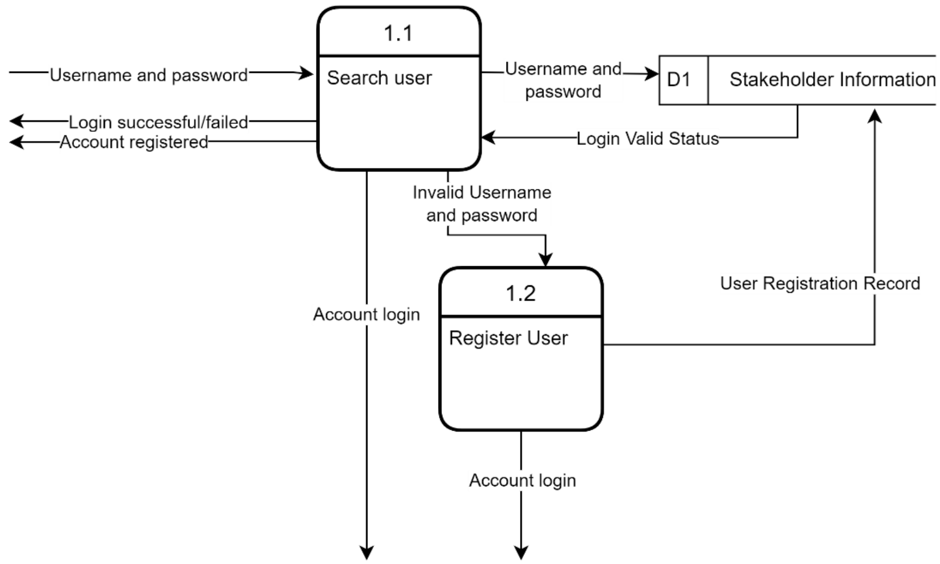
7.1 Physical DFD TO-BE system

7.1.1 Level 0 Diagram

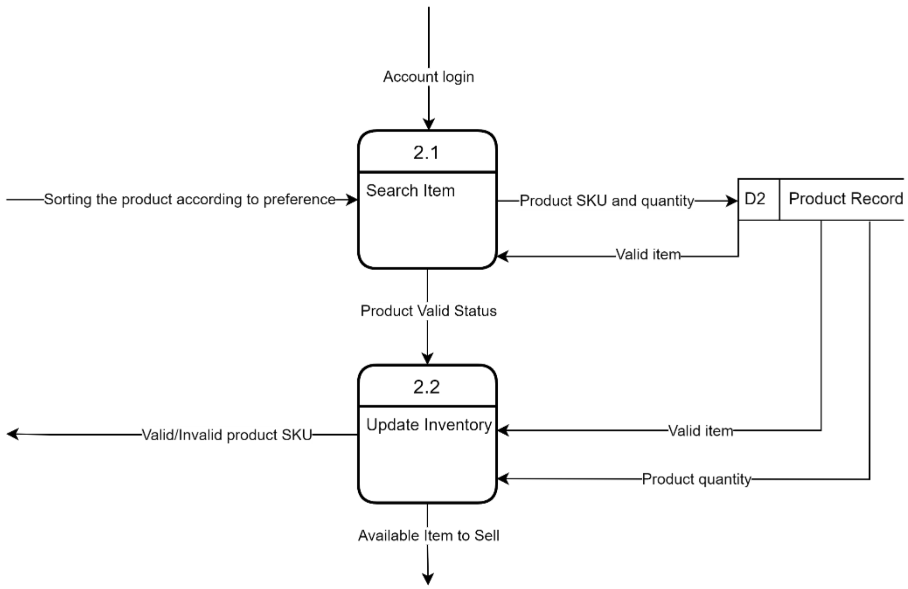


7.1.2 Level 1 Diagram

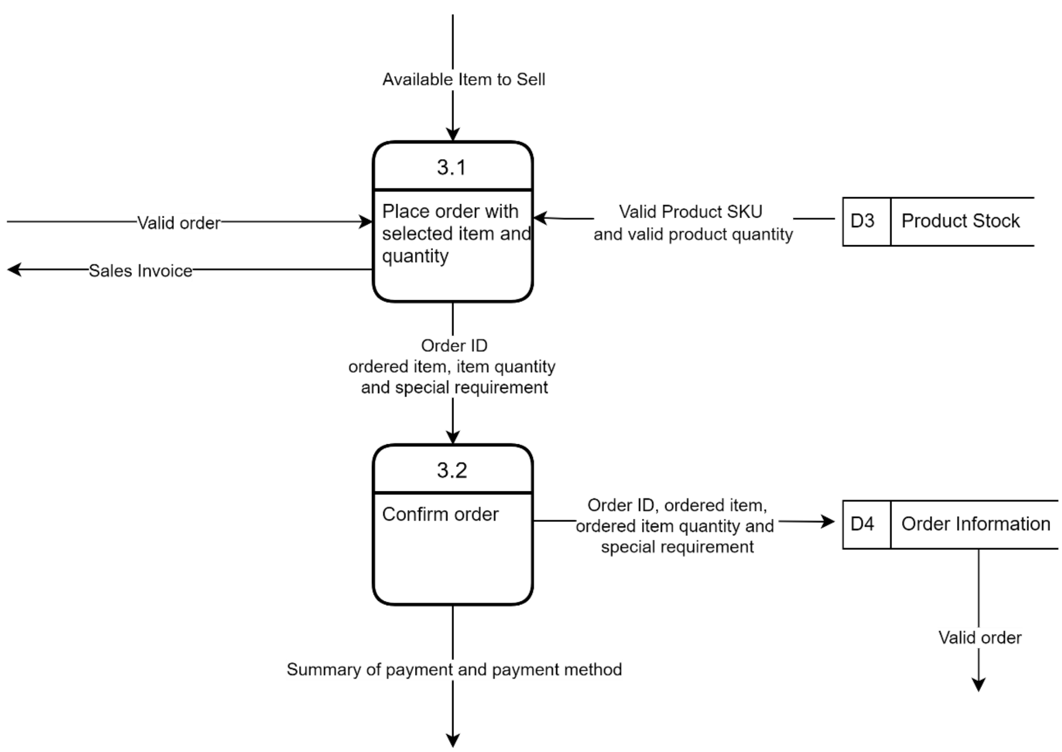
7.1.2.1 Process 1: Login Account



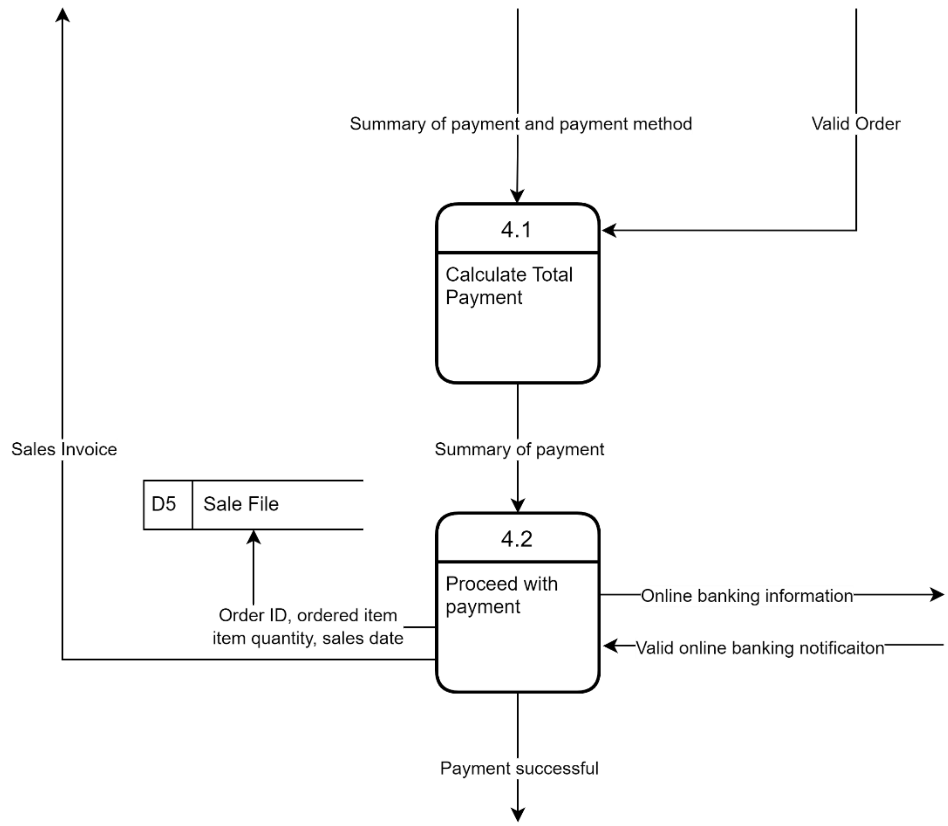
7.1.2.2 Process 2: Check and Update the stock



7.1.2.3 Process 3: Check with order whether available item match with the order



7.1.2.4 Process 4: View Payment status



8.0 Summary of Proposed System

In a nutshell, our group has completed the analysis dan design for Richme Trading. In this phase, we have done the Logical DFD TO-BE system, process specification, Physical DFD-TO-BE system. In the AS-IS system, the system is not fully automated, especially on the inventory part. The TO-BE system is an auto system and makes the sales process become easy. The stakeholder can also check the tracking progress and also make the order analysis. All the information will be automatically stored in the database. Every data has their number to uniquely identify different data such as order has order ID, the package got its own ID and so on. When the users want to check the previously order. The system will allow the user to choose the specific period like month, 3 months or other. After that, the user also can check the profit that earned along the sales process by clicking a button which is easier that the AS-IS system.